

# Break-Aware Weighted Local Linear Forecasting for Cocoa Returns: Nonparametrics vs Machine Learning

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## Abstract

This paper proposes a break-aware weighted local linear (WLL) forecasting framework for cocoa futures returns, combining nonparametric regression with structural break detection and machine-learning benchmarks. We document the empirical performance of WLL relative to random forests and gradient-boosted trees, and we analyze a rare but severe instability of WLL arising from break-aware tuning. We then show how simple regularization and tuning constraints restore the stability predicted by nonparametric theory.[0.5ex] **Keywords:** nonparametric forecasting, structural breaks, local polynomial regression, random forests, gradient boosting.

## 1 Introduction

## 2 Data and Empirical Setting

### 2.1 Cocoa returns and sample construction

### 2.2 Sample splitting and forecasting design

### 2.3 Stylized facts

## 3 Structural Break Detection Framework

### 3.1 Testing procedure

### 3.2 Detected break dates

### 3.3 Implications for forecasting

## 4 Break-Aware WLL Forecasting Method

### 4.1 Baseline local polynomial model

Let  $(x_t, y_t) * t = 1^T$  denote the predictor–response pairs, where  $y_t$  is the (log-)return and  $x_t$  is the predictor (possibly simply time or lagged variables). For a forecast origin  $t_0$ , the local linear

estimator at  $x * t_0$  solves

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_{t \in \mathcal{I}} K_h(x_t - x_{t_0}) (y_t - \beta_0 - \beta_1(x_t - x_{t_0}))^2, \quad (1)$$

with kernel  $K_h(\cdot)$  and index set  $\mathcal{I}$  determined by the training window.

## 4.2 Incorporating break information

Let  $\hat{\tau}$  denote a detected break date within the training window. A break-aware WLL introduces a weight parameter  $\gamma \in [0, 1]$  that governs the relative influence of pre-break and post-break observations.<sup>1</sup>

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<sup>1</sup>Details can be adapted to the final formulation you settle on.



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